

CY 403: Numerical methods and Data Analysis in Chemistry

Course Code	CY 403
Course Name	Numerical methods and Data Analysis in Chemistry
L-T-P-C	3-0-0-3
Prerequisites	Computing and Data Science IC 152
Intended for	BS Chemical Sciences
Mutual Exclusion	None
Approval	42nd Senate (Jan 2024)

Course Contents

- **Theories of Numerical Methods:** (12 Hours)
 - Methods of solution of sets of linear equations: Gauss elimination, Gauss-Jordan elimination, Gauss-Seidel method, QR decomposition method
 - Solution of nonlinear algebraic equations: Bisection method, Newton-Raphson method, Secant method
 - Interpolation and extrapolation: Polynomial interpolation and extrapolation, Rational function interpolation and extrapolation, Cubic spline interpolation
 - Numerical differentiation: Finite-difference method, Higher-order methods
 - Numerical integration: Newton-Cotes quadrature – Rectangle rule, Trapezoidal rule, Simpson's 1/3rd and 3/8th rule, Romberg's method; Gaussian quadrature
 - Solution of differential equations: Euler method, Predictor-corrector method - Improved and Modified Euler method, Runge-Kutta method; Finite-difference method
 - Fourier analysis: Fourier transform of discretely sampled data, Fast Fourier transform
- **Significance and application of the numerical methods in Chemistry:** Application of the solution of the sets of linear equations in Quantum Mechanics, Application of the solution of the nonlinear algebraic equations to get the optimum of the energy landscapes and minimum of the error function, Application of interpolation and extrapolation techniques in Chemistry; to predict data related to chemical experiments at a given condition, Application of the numerical differentiation techniques to solve diffusion equation, Brief introduction to the Molecular Dynamics; Störmer-Verlet, Verlet, Velocity – Verlet methods, Importance of Fourier analysis in spectroscopy. (8 Hours)
- **Data Analysis:** Determining the distribution of a set of data, Moments of a distribution – its mean, variance, skewness etc., Correlation of data – linear correlation, auto-correlation, least square fit method, Importance of data analysis in Chemistry. (4 Hours)
- **Introduction to the Langevin Dynamics Simulations and Monte Carlo Techniques:** Idea of the random numbers, Langevin Dynamics Simulations, Monte Carlo algorithm, Metropolis algorithm, Gillespie algorithm; their significance in Chemistry. (6 Hours)
 - Introduction to time-series analysis and machine learning and their connection to chemistry (12 Hours) Time-series analysis: Trend, stationarity, seasonality and correlations; Moving average (MA), Autoregressive (AR), Autoregressive moving average (ARMA), Autoregressive integrated moving average (ARIMA) models; Forecasting with ARIMA model; Spectral density function and Spectral analysis.
 - Machine learning: Supervised learning and linear regression, Logistic regression, Decision tree and Random forest, Unsupervised learning, Time-series modelling, Deep learning
 - Significance in Chemistry: Illustration of the applicability of time-series analysis and machine learning in important problems related to chemistry; such as calculating bond orders and determining normal modes, forecasting reaction pathways, proposing the designing techniques of new molecules and materials etc.

Text Books

1. H. W. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, **Numerical Recipes - The Art of Scientific Computing**, Cambridge Univ. Press, 1992.
2. F. Jensen, **Introduction to Computational Chemistry**, Second Edition, Wiley, 2006.
3. T. Hastie, R. Tibshirani and J. Friedman, **The Elements of Statistical Learning**, Springer, 2009.
4. R. H. Shumway and D. S. Stoffer, **Time series analysis and its application**, Springer, 2011.

References

1. H. W. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, **Numerical Recipes in Fortran**, Cambridge Univ. Press, 1992.
2. W.H. Press, B.P. Flannery, S.A. Teukolsky, and W.T. Vetterling, **Numerical Recipes in C**, Cambridge Univ. Press, 1990.
3. J. M. Zelle, **Python Programming: An Introduction to Computer Science**, Beedle & Associates, Inc., 2004.
4. Carleo et al., **Machine learning and the physical sciences**, Reviews of Modern Physics, 91, 045002, 2019.